

Deterministic Optimization

Convex Conic
Programming

Introduction

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PSD Cone and SDP

Nonlinear Optimization Modeling

Learning Objectives for this lesson

- Discover one of the most important nonlinear cone: positive semidefinite cone (PSD cone)
- Formulate semidefinite program (SDP) using PSD cone

Positive Semidefinite Matrix

- For a symmetric matrix $A \in R^{n \times n}$, the following are equivalent
 - A is a positive semidefinite matrix
 - the eigenvalues of A are nonnegative
 - $x^T A x \geq 0$ for all $x \in R^n$
 - A can be factored as $A = D^T D$ for some matrix D
- For example:
 - Identity matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ are PSD
 - $\begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} -2 & 1 \\ 1 & 1 \end{bmatrix}$ are not PSD

Positive Semidefinite Cone

- The set of all positive semidefinite matrices in $R^{n \times n}$ is denoted as S_+^n
- S_+^n is a convex cone, called the positive semidefinite cone
 - If $X \in S_+^n$, then all the eigenvalues of αX are α multiples of eigenvalues of X , therefore nonnegative, so αX is PSD, so S_+^n is a cone
 - To show S_+^n is convex:
 - If $X \in S_+^n, Y \in S_+^n$, and $\lambda \in [0,1]$, then
$$x^\top (\lambda X + (1 - \lambda)Y)x = \lambda x^\top Xx + (1 - \lambda)x^\top Yx \geq 0$$

Positive Semidefinite Cone S_+^2

- PSD cone is a high dimensional cone
 - S_+^n has $\frac{(n+1)n}{2}$ elements, i.e. in $\frac{(n+1)n}{2}$ dimensional space
 - The simplest S_+^2 is already in 3 dimensional space. This is the only PSD cone we can visualize

Linear Constraint of Matrices

- A matrix $A \in R^{n \times n}$ can be viewed as a long vector in R^{n^2} by stacking the columns of A on top of each other:

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} \quad \Rightarrow \quad \text{vec}(A) = \begin{bmatrix} 2 \\ 1 \\ 3 \\ 4 \end{bmatrix}$$

- Linear constraint in matrix can be defined through vectors made out of matrices
 - $\langle A, B \rangle = \text{vec}(A)^\top \text{vec}(B) = \sum_{i=1}^n \sum_{j=1}^n A_{ij} B_{ij}$

Positive Semidefinite Constraint

- One linear matrix constraint in a PSD matrix:

$$\langle A, X \rangle = b, \quad X \in S_+^n$$

- Multiple linear matrix equalities in a PSD matrix:

$$\langle A_1, X \rangle = b_1$$

$$\langle A_2, X \rangle = b_2$$

...

$$\langle A_m, X \rangle = b_m$$

$$X \in S_+^n$$

Semidefinite Program

- A semidefinite program can be written as

$$\begin{array}{ll}\min & \langle C, X \rangle \\ \text{s.t.} & \langle A_1, X \rangle = b_1 \\ & \langle A_2, X \rangle = b_2 \\ & \dots \\ & \langle A_m, X \rangle = b_m \\ & X \in S_+^n\end{array}$$

Summary

- We learned:
 - What is a PSD matrix
 - What is a linear matrix equality
 - What is a SDP